



PORT ARTHUR REFINERY

MOTIVA BOILER COMPLEX

PIPING & INSTRUMENTATION DIAGRAMS

DWG. NO.	TITLE
0000-PID-0001	PAMC TO BOILER COMPLEX
0000-PID-0002	AROMATICS TO BOILER COMPLEX
0000-PID-0003	RES7 TO BOILER COMPLEX
0000-PID-0010	BOILER COMPLEX TO PAMC
0000-PID-0011	BOILER COMPLEX TO AROMATICS
0000-PID-0020	COOLING WATER DISTRIBUTION
0000-PID-0030	NITROGEN DISTRIBUTION
0000-PID-0040	INSTRUMENT AIR DISTRIBUTION
1000-PID-0001	SAND FILTER A
1000-PID-0003	SAND FILTER B
1000-PID-0005	SAND FILTER C
1000-PID-0011	FILTERED WATER
1000-PID-0013	SUPERNATANT SUMP
1010-PID-0001	SODIUM BISULFITE INJECTION SYSTEM
1010-PID-0003	CATION EXCHANGER A
1010-PID-0005	CATION EXCHANGER B
1010-PID-0007	CATION EXCHANGER C
1010-PID-0009	DEMIN RINSE RECYCLE PUMPS
1010-PID-0011	ACID DILUTION SYSTEM
1010-PID-0013	ANION EXCHANGER A
1010-PID-0015	ANION EXCHANGER B
1010-PID-0017	ANION EXCHANGER C
1010-PID-0019	CAUSTIC DILUTION HEATER
1010-PID-0021	CAUSTIC DILUTION SYSTEM
1010-PID-0023	MIXED BED EXCHANGER A
1010-PID-0025	MIXED BED EXCHANGER B
1010-PID-0027	MIXED BED EXCHANGER C
1010-PID-0029	MIXED BED AIR SCOUR BLOWERS
1010-PID-0031	BRINE SYSTEM
1010-PID-0033	DEMIN WATER TANK
1020-PID-0001	DEAERATOR SAMPLE SYSTEM
1020-PID-0003	DEAERATOR A
1020-PID-0005	DEAERATOR B
1020-PID-0007	BFW SAMPLE SYSTEM
1020-PID-0009	OXYGEN SCAVENGER PACKAGE
1020-PID-0011	NEUTRALIZING AMINE PACKAGE
1020-PID-0013	BFW PUMPS A/B
1020-PID-0015	BFW PUMPS C/D
1100-PID-0001	FUEL GAS COALESCER
1100-PID-0002	BFW AND HP STEAM BOILER A
1100-PID-0006	HP STEAM HEADER
1100-PID-0007	MP STEAM HEADER

PIPING & INSTRUMENTATION DIAGRAMS

DWG. NO.	TITLE
1100-PID-0008	LP STEAM HEADER
1100-PID-0009	LP STEAM VENT CONDENSER
1100-PID-0010	BOILER A HP BOILER FEED WATER HEADER
1100-PID-0011	BOILER A EXTERNAL AND INTERNAL ECONOMIZER
1100-PID-0012	BOILER A STEAM DRUM
1100-PID-0013	BOILER A SUPERHEATER
1100-PID-0014	BOILER A FORCED DRAFT FAN
1100-PID-0015	BOILER A STACK
1100-PID-0016	BOILER A AFCU
1100-PID-0017	BOILER A SCR
1100-PID-0018	BOILER A FUEL GAS AND INSTRUMENT HEADERS
1100-PID-0019	BOILER A BURNERS
1100-PID-0020	BOILER A FD FAN LUBE OIL SYSTEM
1100-PID-0021	BOILER A FD FAN STEAM TURBINE
1100-PID-0022	BOILER A BLOWDOWN WATER AND STEAM SAMPLING
1100-PID-0071	PHOSPHATE PACKAGE
1100-PID-0073	PHOSPHATE PACKAGE
1100-PID-0075	CONTINUOUS BLOWDOWN DRUM
1100-PID-0077	INTERMITTENT BLOWDOWN DRUM
1100-PID-0079	BLOWDOWN PUMPS AND COOLER
1100-PID-0083	HP TO MP DESUPERHEATER
1100-PID-0087	MP TO LP DESUPERHEATER
1500-PID-0001	CONDENSATE FLASH DRUM A
1500-PID-0007	CONDENSATE EXCHANGERS
1500-PID-0011	CONDENSATE TRIM COOLERS
1500-PID-0015	CONDENSATE TANK
1500-PID-0017	CONDENSATE PUMPS
1500-PID-0019	ACTIVATED CARBON FILTERS
1500-PID-0021	CONDENSATE POLISHER A
1500-PID-0023	MIXED BED POLISHER B
1500-PID-0031	CONDENSATE CAUSTIC DILUTION HEATER
1500-PID-0033	CAUSTIC DOSING PUMPS
1500-PID-0035	SULFURIC ACID DOSING PUMPS
2000-PID-0001	AMMONIA TANK A
2000-PID-0003	AMMONIA TANK B
2000-PID-0005	SODIUM HYDROXIDE STORAGE
2000-PID-0007	SULFURIC ACID TANK
3000-PID-0001	STORMWATER SUMP
3000-PID-0003	NONOILY WATER COLLECTION
3000-PID-0005	OILY WATER COLLECTION
3000-PID-0007	SANITARY LIFT STATION

PIPING & INSTRUMENTATION DIAGRAMS

DWG. NO.	TITLE
YC-169496	DRINKING WATER UTILITY MAP
YC-199790	PC STORMWATER HOLDING TANK
YC-199791	PC STORMWATER HOLDING TANK TRANSFER/ RECOVERED OIL PUMPS
YC-216699	POTABLE WATER DISTRIBUTION
YC-216702	INSTRUMENT AIR DISTRIBUTION
YC-216703	NITROGEN DISTRIBUTION
YC-216710	190 PSIG (LP) NATURAL GAS DISTRIBUTION
YC-216712	REFINERY FUEL GAS DISTRIBUTION
YC-216717	PROCESS WASTE WATER COLLECTION
YC-216983	CLARIFIED WATER PUMPS

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND CO-ORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	01-09-2026	ISSUED FOR FEL2B REPORT	SSP	GS							

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

COVER SHEET

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02

DRAWN BY: SSP

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: GS

ENG APVL:

25MA02-PRJ-100

REV A